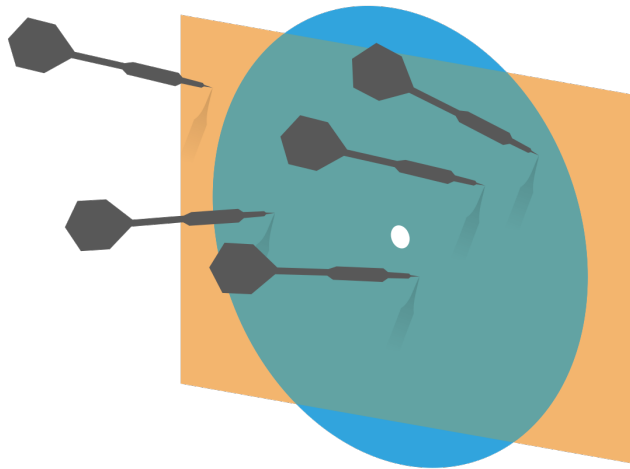


Probability and Statistics for Computer Science



“A major use of probability in statistical inference is the updating of probabilities when certain events are observed” – Prof. M.H. DeGroot

Credit: wikipedia

PollEv site



Pollev.com/hongyeliu713

Last time

Probability a first look

- ✱ Outcome and Sample Space
- ✱ Event
- ✱ Probability: Axioms & Properties
- ✱ Calculating probability

Objectives

✱ Probability

- ✱ Probability calculation

- ✱ Conditional Probability

 - ✱ Bayes rule

 - ✱ Independence

Warm up

We have 4 brands of milk to give 3 babies, how many outcomes are there the babies get different milk?

$$4 \times 3 \times 2 = 24$$

Warm up (II)

work at home

✻ Fill the blanks:

“I am an avid vegetarian and I enjoy eating all day long, people admire my appetite and like to watch me eat.” I am a _____. How many ways are there to rearrange these 5 letters? _____. If you draw 2 letters from them, how many outcomes (order matters) are there that are without “a”?

Probability by reasoning with the complement property

✱ If $P(E^c)$ is easier to calculate

$$P(E) = 1 - P(E^c)$$

Probability by reasoning with the complement property

- ✱ A person is taking a test with **N** true or false questions, and the chance he/she answers any question right is 50%, what's probability the person answers **at least** one question right?

E : ?

E^c : None is right

$$P(E) = 1 - P(E^c)$$

$$|E| = 1$$

$$|\Omega| = 2 \times 2 \times \dots \times 2 = 2^N$$

$$\left(1 - \left(\frac{1}{2}\right)\right)^N$$

Probability by reasoning with the union property

✱ If E is either E_1 or E_2

$$P(E) = P(E_1 \cup E_2) =$$

$$P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

Probability by reasoning with the properties (2)

- ✱ A person may ride a bike on any day of the year equally. What's the probability that he/she rides on a Sunday or on 15th of a month?

$E: ?$

$$P(\text{Sun}) = \frac{52}{366}$$

$$P(15^{th}) = \frac{|E| = 12}{366}$$

$$P(\text{Sun} \cap 15^{th}) = \frac{2}{366}$$

(15th & Sun)
Sept 15th, Dec 15th

$$\begin{aligned} P(E) &= P(\text{Sun} \cup 15^{th}) \\ &= \frac{52}{366} + \frac{12}{366} - \frac{2}{366} \end{aligned}$$

Senate Committee problem

The United States Senate contains **two** senators from each of the **50** states. If a committee of eight senators is selected at random, what is the probability that it will contain at least one of the two senators from **IL**?

E^c : None of two IL sen

$$P(E^c) = \frac{|E^c|}{|S|} = \frac{\binom{98}{8}}{\binom{100}{8}}$$

$$P(E) = 1 - P(E^c)$$

Probability: Birthday problem

- ✱ Among 30 people, what is the probability that at least 2 of them celebrate their birthday on the same day? Assume that there is no February 29 and each day of the year is equally likely to be a birthday.

E^c : None of them share B-day

$$\frac{|E^c|}{|\Omega|} = \frac{365 \times 364 \times \dots \times 335}{365^{30}}$$

What if the event has infinite outcomes

✱ Tossing a fair coin until head appears

✱ Coin is tossed at least 3 times

This event includes infinite # of outcomes.

And the outcomes don't have equal probability.

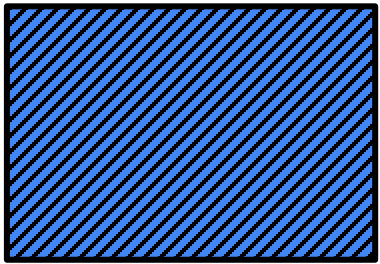
$$1 - P(E^c)$$

TTH, TTTH, TTTTH....

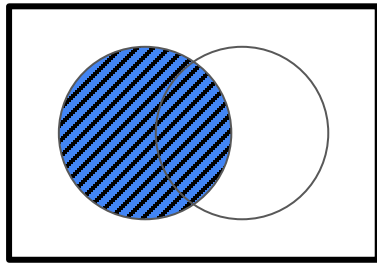
$$E^c = \{H, TH\}$$

$$P(TH) = P(\overset{1st}{T} \& \overset{2nd}{H})$$

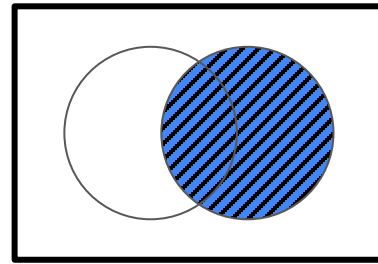
Venn Diagrams of events as sets



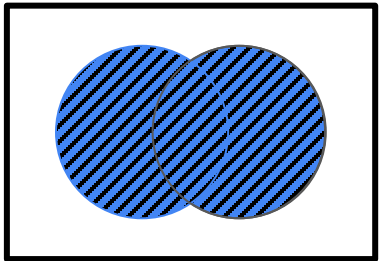
Ω



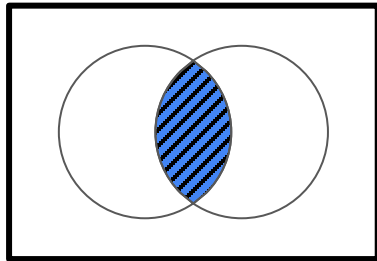
E_1



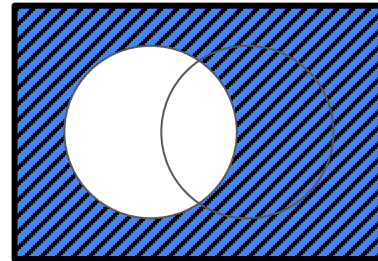
E_2



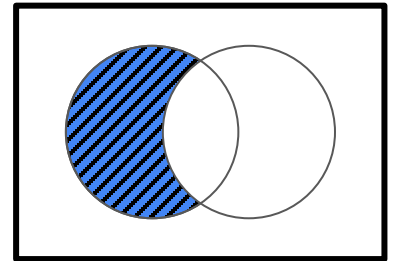
$E_1 \cup E_2$



$E_1 \cap E_2$



E_1^c



$E_1 - E_2$

Conditional Probability

✱ Motivation of conditional probability

Conditional Probability

✱ Example:

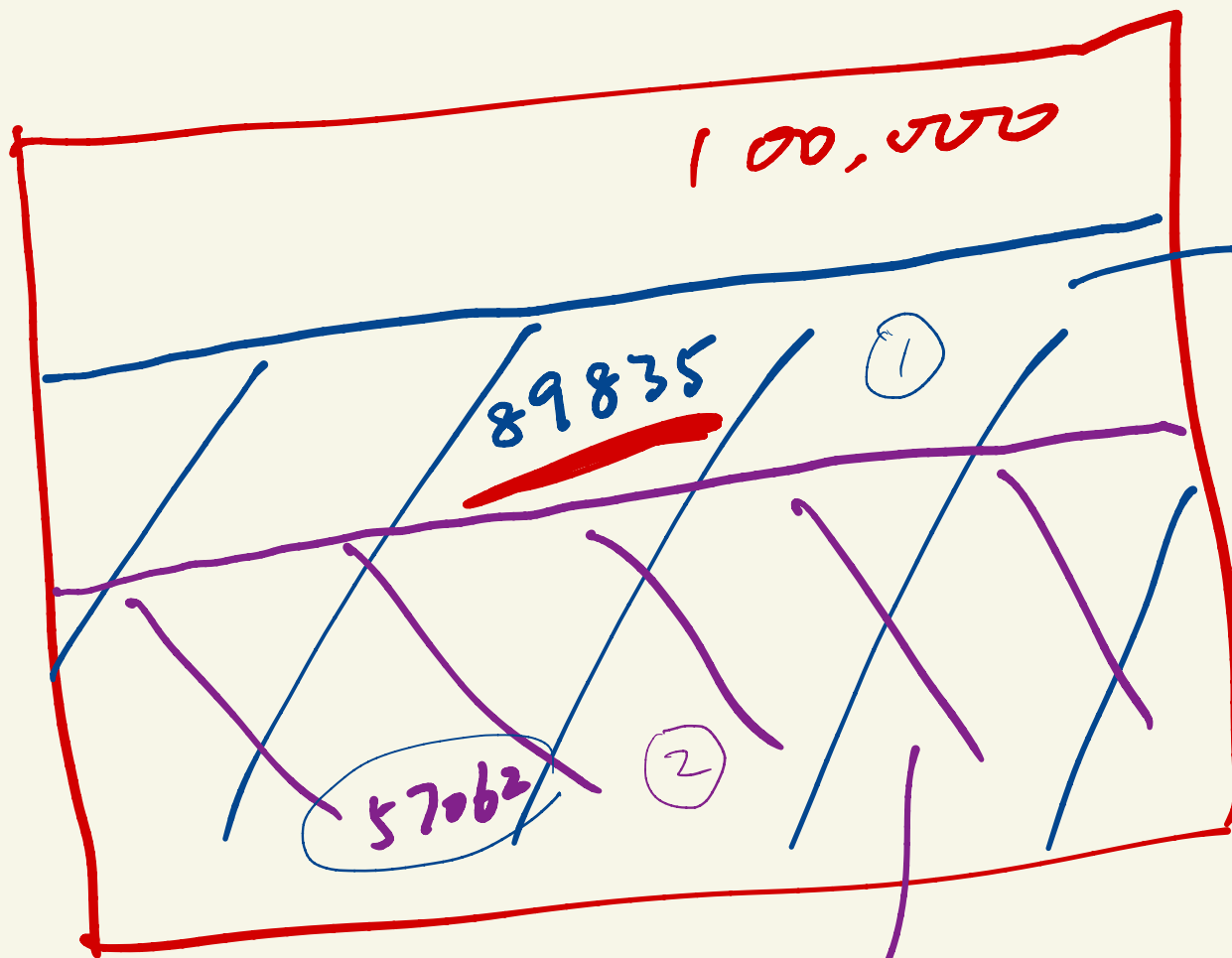
An insurance company knows in a population of 100 thousands females, 89.835% expect to live to age 60, while 57.062% can expect to live to 80. Given a woman at the age of 60, what is the probability that she lives to 80?

Conditional Probability

✱ Given the condition she is [≥]60 already, the size of the sample space for the outcomes has changed to

89,835 instead of 100,000

$$|\Omega| = 100,000$$



live to ≥ 60

$$|\bar{E}_{60}| =$$

$$|\bar{E}_{80}| =$$

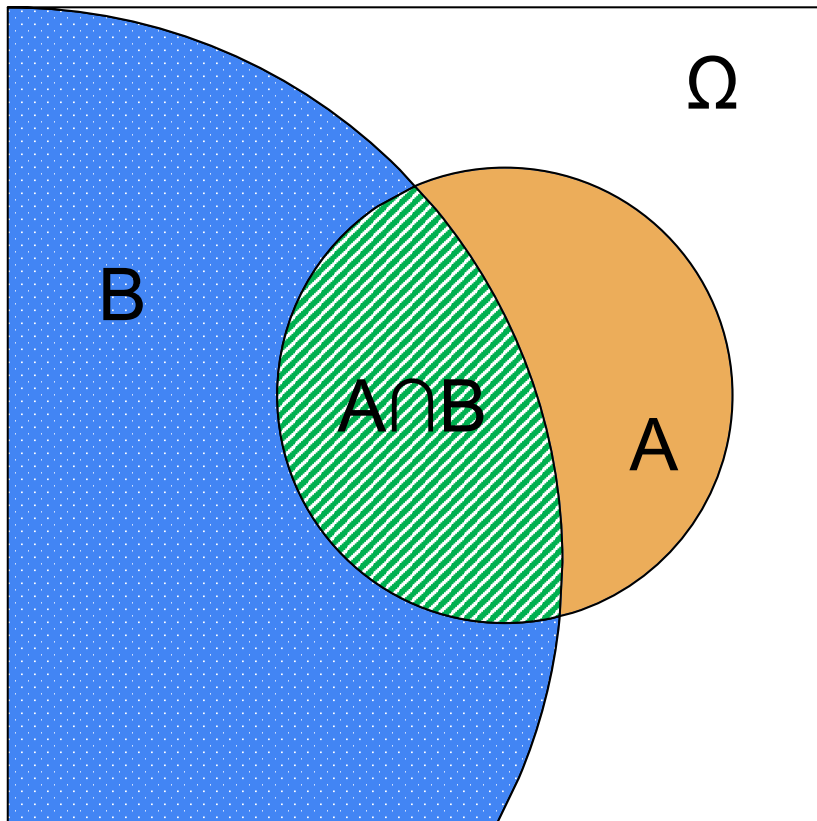
$$P(\bar{E}_{80} | \bar{E}_{60}) = \frac{|\bar{E}_{80}|}{|\bar{E}_{60}|} = \frac{57062}{89835}$$

live to 80

$$|\Omega_n| = ? \quad 89835$$

Conditional Probability

✱ The probability of **A** given **B**



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(B) \neq 0$ *↳ prior*

The “Size” analogy

Conditional Probability

A : a woman
lives to at least
80

$$P(A|B) = \frac{57,062}{89,835} = 0.6352$$

B : a woman is
at least 60 now

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$= \frac{57,062/100,000}{89,835/100,000}$$

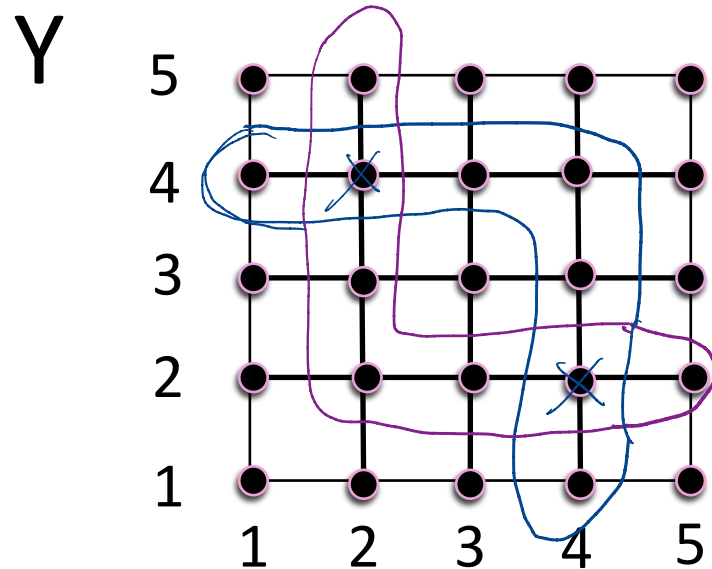
While $P(A) = \frac{57,062}{100,000} = 0.57062$

Conditional Probability: die example

Throw 5-sided fair die twice.

$$A : \max(X, Y) = 4$$

$$B : \min(X, Y) = 2$$



X

$$P(A|B) = ? \quad \frac{P(A \cap B)}{P(B)} = \frac{2/25}{7/25} = \frac{2}{7}$$

Conditional probability, that is?

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

a probability!

→ Three axioms

$$P(A|B) + P(A^c|B) = ? \quad \textcircled{1}$$

$$\frac{P(A \cap B)}{P(B)} + \frac{P(A^c \cap B)}{P(B)} = \frac{P(B)}{P(B)}$$

$$P(A_1 \cup A_2|B) = ? \quad \text{if } A_1 \cap A_2 = \emptyset$$

$$= P(A_1|B) + P(A_2|B)$$

Multiplication rule using conditional probability

✱ Joint event

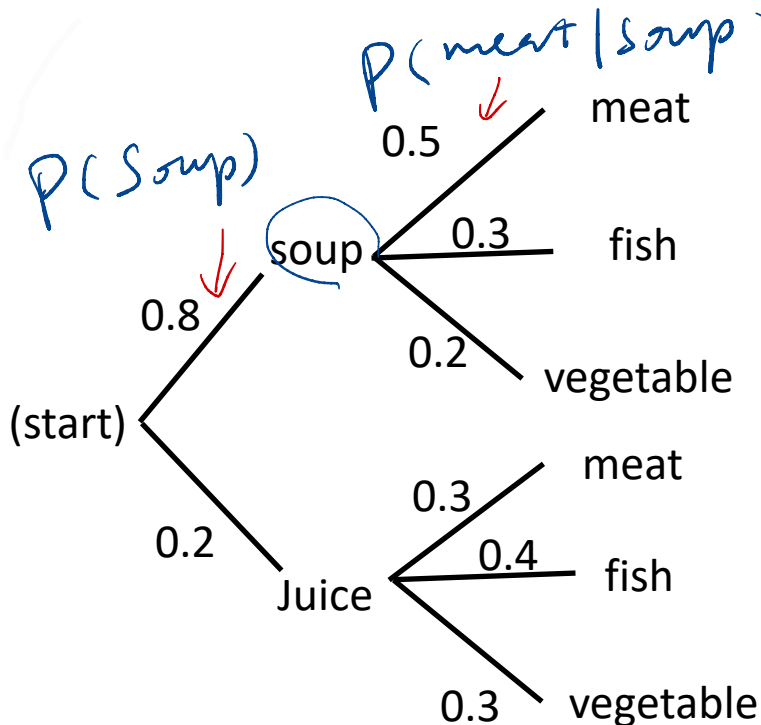
$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

$$\Rightarrow P(A \cap B) = P(A|B)P(B)$$

↓
prior

Multiplication using conditional probability

$$P(A \cap B) = P(A|B)P(B)$$

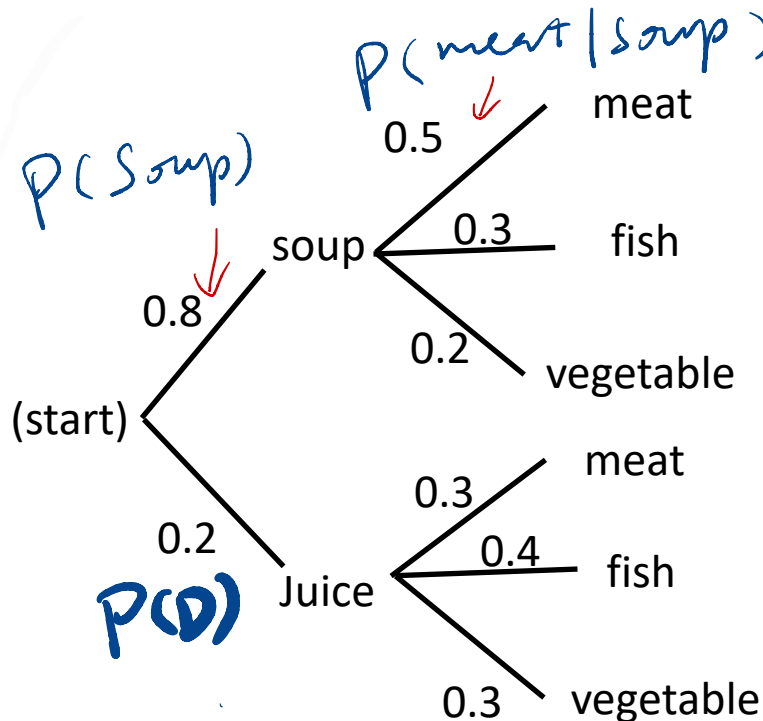


$$P(\text{soup} \cap \text{meat}) = P(A|B)P(B) \\ = 0.4 \quad = 0.5 \times 0.8$$

$$P(\text{soup} \cap \overline{\text{fish}}) \\ = P(\text{soup} \cap \text{meat}) + P(\text{soup} \cap \text{veg}) \\ = 0.5 \times 0.8 + 0.8 \times 0.2$$

Multiplication using conditional probability

$$P(A \cap B) = P(A|B)P(B)$$



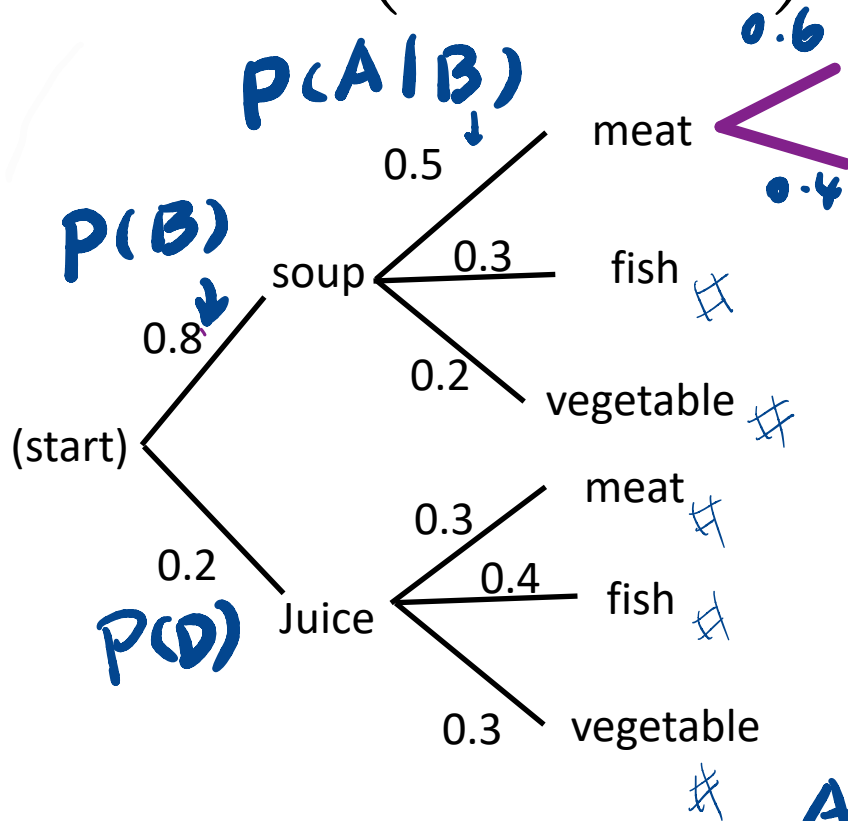
$$P(A^c|B) = ? \quad 1 - P(A|B)$$
$$P(A^c \cap D) = ? \quad 1 - \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} &\downarrow \\ &0.8 \\ &= 1 - \frac{0.4}{0.8} \\ &= 0.5 \end{aligned}$$

A : meat B : soup
 D : juice

Multiplication using conditional probability

$$P(A \cap B) = P(A|B)P(B)$$



icecream
cheesecake

C: cheesecake

$$P(C|B) = ?$$

A: meat B: soup D: juice

Symmetry of joint event in terms of conditional prob.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

$$\Rightarrow P(A \cap B) = P(A|B)P(B)$$

$$\Rightarrow P(B \cap A) = P(B|A)P(A)$$

Symmetry of joint event in terms of conditional prob.

$$\because P(B \cap A) = P(A \cap B)$$

$\Rightarrow$

$$P(A|B)P(B) = P(B|A)P(A)$$

The famous Bayes rule

$$P(A|B)P(B) = P(B|A)P(A)$$

$$\Rightarrow P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)}$$



Thomas Bayes (1701-1761)

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

Assignments

- ✱ Work on Module Week 3 on Canvas
- ✱ Next time: More on independence and conditional probability

See you next time

See You!

